

Machine Learning as a Future Direction for Data Analysis in Myofilament Biomechanics

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Introduction

Kinesiology is a multifaceted discipline seeking to understand the mechanics of human movement and physiology. As the field has evolved, the hardware utilized to measure these mechanics, such as high-fidelity force transducers and dynamic length controllers, has become remarkably sophisticated. However, the computational frameworks used to interpret these measurements lag significantly behind the physical equipment. In the sub-discipline of muscle biomechanics, researchers assessing active and passive stress frequently encounter a severe computational bottleneck. Currently, researchers must manually extract force data by visually approximating complex, high-frequency graphical traces. Approaching this challenge with a mindset oriented toward advanced informatics, it is evident this analytical workflow must be modernized. The transition from manual data extraction to automated machine learning (ML) pipelines for identifying stress-relaxation plateaus, calculating active stress, and generating sigmoidal regressions represents a critical future direction in kinesiology. By leveraging pattern recognition, mechanical assessments can be extracted objectively, matching the throughput of modern laboratory hardware (Aboelkassem, 2025; Lee et al., 2025).

The Current State of the Field

Presently, the assessment of skeletal and cardiac muscle mechanics relies heavily on laboratory procedures evaluating force production and passive compliance. Researchers frequently employ active-passive testing protocols where isolated muscle preparations, such as chemically skinned cardiac trabeculae, are stretched and sequentially activated. During these tests, the physical equipment operates at extremely high temporal resolutions. Force transducers routinely sample data at frequencies exceeding 1000 Hertz to capture the millisecond-to-millisecond details of a muscle fiber's response to rapid shortening or calcium activation.

While the physical hardware is exceptional, the subsequent data management workflows are antiquated. A standard protocol can generate hundreds of thousands of data points per minute. Although these high-frequency sampling rates allow the experiment to be exported as massive comma-separated values (CSV) files, legacy spreadsheet software cannot efficiently process these immense text files without

severe performance degradation. Consequently, researchers bypass the raw data, physically monitoring the testing terminal to manually record specific data points of interest from a live graphical interface.

Extracting mechanical parameters via visual inspection is computationally burdensome and highly susceptible to human error. To determine passive stress, muscles are stretched to specific sarcomere lengths and held constant. The researcher must visually identify the exact moment the exponential decay curve flattens into a steady-state plateau. Evaluating calcium sensitivity requires repeating this extraction across sequentially increasing calcium concentrations to map a force-pCa curve. Danieli-Betto et al. established a foundational approach for calcium sensitivity analysis in skinned skeletal muscle fibers, including pCa-based force assessment (Danieli-Betto et al., 1990). However, the standard workflow still relies heavily on steady-state assumptions and human judgment when identifying the relevant region of the trace. The researcher must then use external software to fit a non-linear sigmoidal regression, the Hill equation, to calculate the pCa50. Across these protocols, cognitive fatigue drastically increases the likelihood of human error, forcing researchers to base delicate molecular analyses on subjective approximations rather than mathematical rigor.

Future Developments and Emerging Directions

The future of data analysis in kinesiology lies in applying machine learning and automated signal-processing methods to overcome manual signal extraction and unlock the full utility of high-frequency datasets. Although absolute force values vary across biological samples, the underlying signal morphology, including stress-relaxation decay patterns and steady-state force plateaus, remains sufficiently consistent to support automated feature extraction.

The feasibility of this approach is supported by recent successes in macroscopic exercise physiology, specifically the automated analysis of breath-by-breath oxygen uptake (VO_2) kinetics. Machine learning models have been used to detect non-linear physiological inflection points, such as ventilatory thresholds, with expert-level accuracy (Zignoli et al., 2019). The computational challenge in VO_2 analysis, filtering biological noise to identify a physiologically meaningful threshold, is mathematically similar to the challenge of detecting active and passive stress features in isolated mus-

cle fiber recordings. However, critically evaluating Zignoli et al.'s methodology reveals a distinct limitation. Their recurrent neural network model was trained on a database of 228 cardiopulmonary exercise tests, which highlights the importance of substantial labeled training data (Zignoli et al., 2019). A significant hurdle for adapting this to myofilament mechanics is the comparative scarcity of standardized, open-access trabeculae datasets, likely necessitating the use of synthetic data or transfer learning to train initial force-extraction models.

Building on this precedent, future muscle biomechanics laboratories will employ time-series forecasting models and dynamic algorithms to bypass manual extraction. For example, a computer program utilizing a sliding window algorithm can continuously calculate the derivative, or slope, of the incoming force data, objectively registering a steady-state plateau when the slope reaches zero and remains within a defined range. For more complex pattern recognition during active contractions, researchers will utilize one-dimensional convolutional neural networks (1D-CNNs). Machine learning is also being integrated more broadly with physics-based modeling of physiological systems to improve computational efficiency and parameter estimation (Lee et al., 2025). Specifically, Aboelkassem developed a deep learning model of myofilament cooperative activation and cross-bridge cycling in cardiac muscle, showing that machine learning can predict steady-state cardiac sarcomere contraction behavior from model-generated data (Aboelkassem, 2025). However, a limitation in extending such modeling to pathological tissue is that local structural heterogeneity, such as fibrosis or collagen deposition, may violate simplified assumptions of spatial uniformity.

To overcome these limitations, a custom machine learning pipeline acting as modern middleware is required. This software would automatically ingest full-resolution CSV files, mirroring the efficiency gains reported in AI-powered skeletal muscle ultrasonography. Rivera et al. reported that their automated software reduced lower-limb muscle ultrasound analysis time from 24 hours to 247 seconds while maintaining strong comparability with human analysis (Rivera et al., 2025). While Rivera et al. used image-based models rather than force-trace models, adapting that broader automation logic to one-dimensional time-series force data provides a plausible path forward. The algorithm would pinpoint the stress-relaxation steady-state, determine pure active stress, and rapidly generate the sigmoidal regressions required to identify the pCa50.

Implications for Kinesiology

The integration of automated data pipelines carries profound implications for kinesiology research and systemic interoperability. Primarily, this development resolves a pervasive operational bottleneck. By relieving the immense time

constraints associated with monitoring traces and logging parameters, researchers are afforded significantly more time for rigorous experimental design and the translation of findings. Furthermore, this shift directly enhances the field's reproducibility. Human analysis of graphical plateaus carries an inherent risk of subconscious confirmation bias. An appropriately trained ML model processes data objectively based solely on its mathematical parameters, virtually eliminating inter-rater variability.

However, AI-driven data extraction also carries clear computational risks. Overfitting can cause a model to perform well on training data but poorly on new recordings, while corrupted signals, mislabeled training examples, and mechanical artifacts can produce confident but physiologically invalid outputs. In myofilament testing, one relevant example is the possibility that a transient disturbance could be misclassified as a meaningful contractile event if the model is not properly constrained. More broadly, current work in computational physiology emphasizes that machine learning systems are most useful when integrated carefully with mechanistic understanding rather than treated as isolated black-box predictors (Lee et al., 2025). To reduce this risk, the automated pipeline must implement data gating and transient filtering. The algorithm must be programmed to apply a mathematical threshold that automatically flags non-physiological rates of force development, such as a tension spike occurring faster than biological cross-bridge cycling allows.

Furthermore, utilizing purely autonomous black-box algorithms prevents researchers from explaining how a statistical conclusion was reached. To mitigate these methodological flaws, future computer applications in kinesiology should incorporate interpretable outputs within intuitive user interfaces and support a strict human-in-the-loop paradigm (Lee et al., 2025). Artificial intelligence should serve as a high-speed assistant, not a replacement for physiological expertise. The system must present the automated analysis, display the algorithm's confidence score, and visually highlight the selected plateaus for the researcher to rapidly verify or reject. This synergistic approach ensures high-throughput data processing without sacrificing scientific rigor.

Conclusion

The current state of data analysis in myofilament biomechanics relies heavily on the manual, time-consuming extraction of active stress, passive stress, and calcium sensitivity from visual approximations of live traces. This antiquated methodology abandons high-resolution data, limits the speed of scientific discovery, and introduces subjective bias. By applying evidence-informed machine learning techniques to full-resolution, high-frequency CSV outputs, the field of kinesiology can automate the identification of stress-relaxation plateaus and the generation of sigmoidal regressions. Supported by precedents in macroscopic exercise

physiology and recent work in machine learning for cardiac sarcomere modeling and automated musculoskeletal imaging, this future direction is increasingly feasible and scientifically important (Aboelkassem, 2025; Lee et al., 2025; Rivera et al., 2025; Zignoli et al., 2019). Ultimately, streamlining this data extraction process will solve a persistent operational bottleneck, allowing researchers to evaluate muscle mechanics with greater objectivity, standardization, and efficiency.

AI Disclosure

In accordance with course policies, the grammatical enhancement and formatting of this paper were assisted by an AI language model.

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